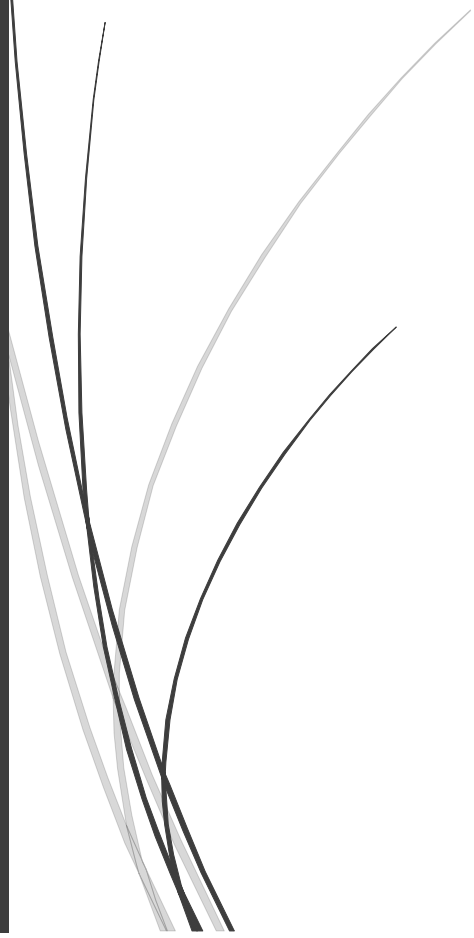
N+ -Network Security

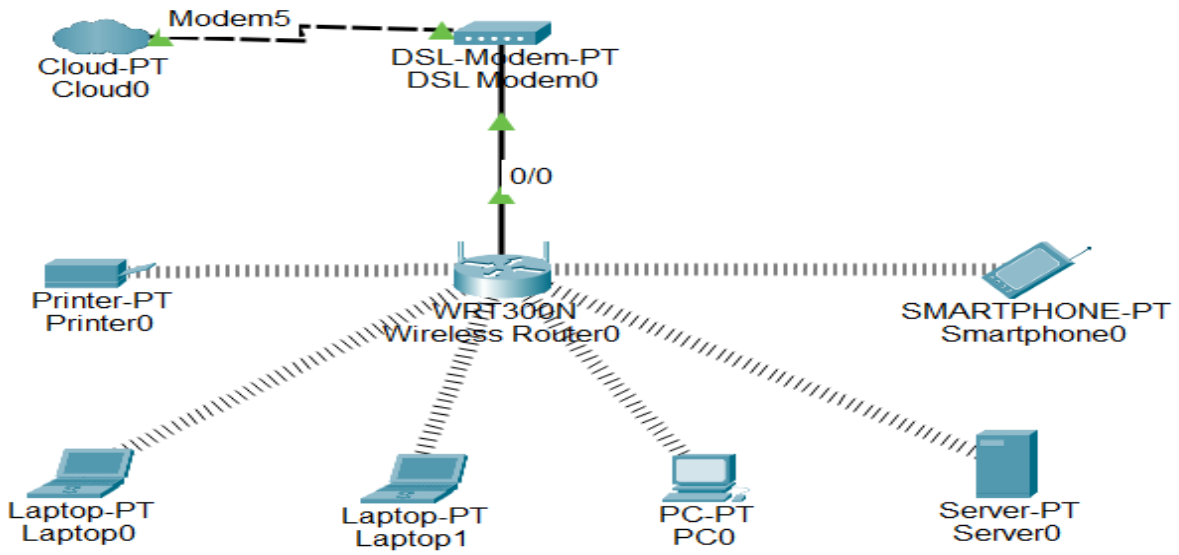
10/24/2026

Assignment-1



Prepare By
~PINTU MER

1. **Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).**



2. **List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed. Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.**

#	Device / Component	Function in Topology	Recommended Brand & Model (Indian Market)	Approx. Cost (INR)
1	ISP Terminal / Cloud Gateway	Provides incoming Internet Service Provider (ISP) line connection.	Fiber ONU / ONT • *Digisol DG-GR1321* • *Syrotech GPON ONT*	₹1,200 – ₹1,800
2	DSL Modem	Converts raw incoming DSL/telephone line signals to Ethernet data.	DSL Broadband Modem • *TP-Link TD-W8961N* • *D-Link DSL-2750U*	₹1,200 – ₹2,000
3	Wireless Router	Handles network routing, DHCP IP distribution, NAT firewall, and Wi-Fi signal.	Dual-Band Wi-Fi Router • *TP-Link Archer C6 AC1200* • *D-Link R15 AX1500*	₹2,200 – ₹3,500
4	Telephone Cable (RJ-11)	Connects the incoming wall jack / ISP line to the DSL Modem line port.	Standard RJ-11 Phone Cable	₹100 – ₹150

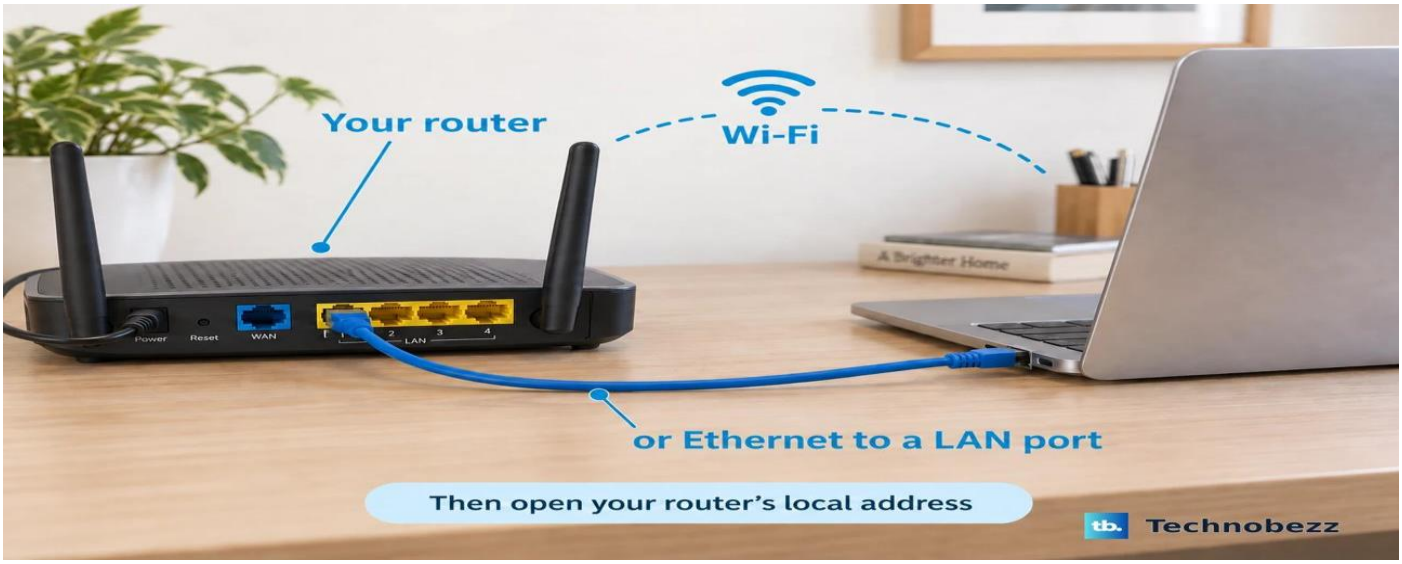
#	Device / Component	Function in Topology	Recommended Brand & Model (Indian Market)	Approx. Cost (INR)
5	Ethernet Cables (RJ-45)	Connects DSL Modem to Router WAN port and connects wired PCs.	Cat6 Patch Cord • *D-Link / FEDUS Cat6 Cable*	₹150 – ₹300
6	Client Devices	Laptops, PCs, or mobile devices connecting via wireless or wired network.	• Built-in Wi-Fi / USB Wi-Fi Adapters (*TP-Link TL-WN725N*)	N/A
7	Ethernet Switch	Expands wired connections if you have more than 4 wired PCs.	TP-Link TL-SG105 (5-Port Gigabit)	₹1,100
8	Wi-Fi Extender	Boosts wireless range to eliminate Wi-Fi dead zones.	TP-Link RE305 (AC1200 Extender)	₹2,100

3. Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

SOHO Wi-Fi Router Configuration – Step by Step

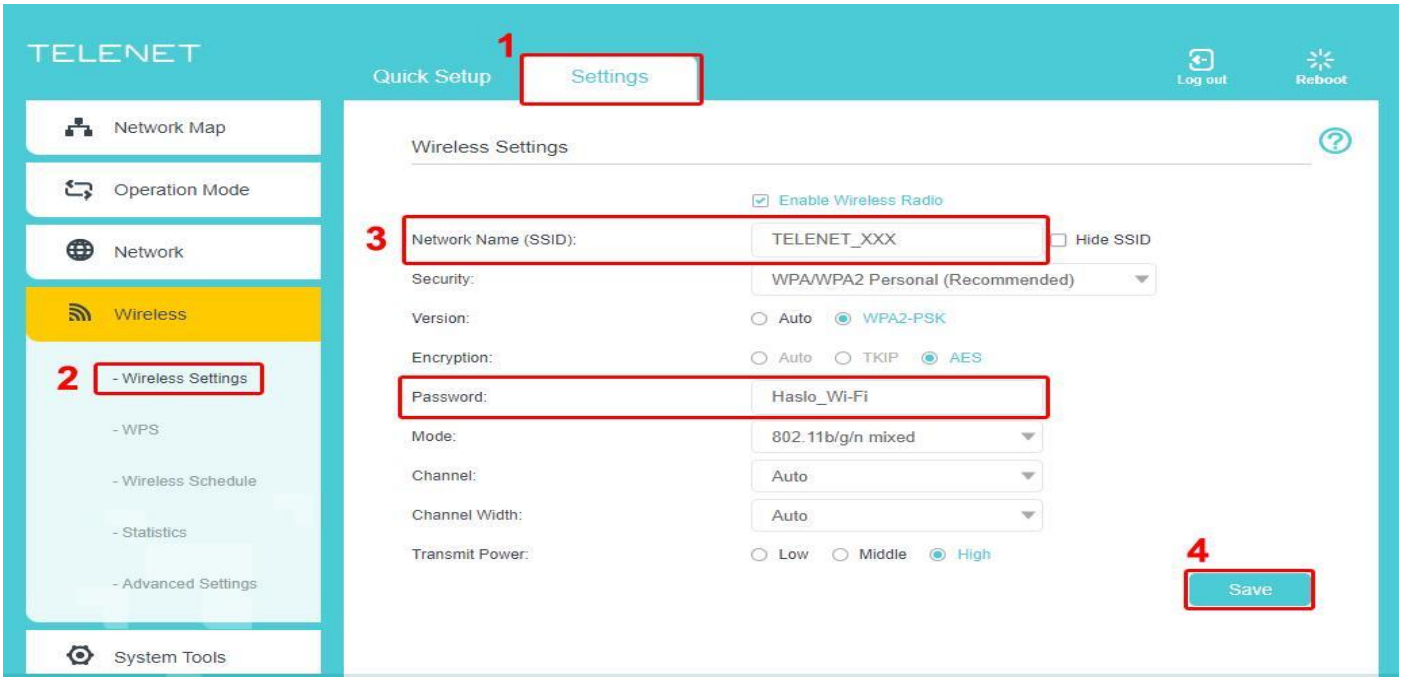
Aim: To configure a Wi-Fi router for a SOHO (Small Office/Home Office) network by setting a unique SSID, enabling WPA2 security, and changing the default admin password.

Step 1: Connect to the Wi-Fi Router



1. Connect your PC or laptop to the router using an Ethernet cable or Wi-Fi.
2. Open a web browser.
3. Enter the router's IP address, such as 192.168.1.1 or 192.168.0.1.
4. Log in using the router's admin credentials.

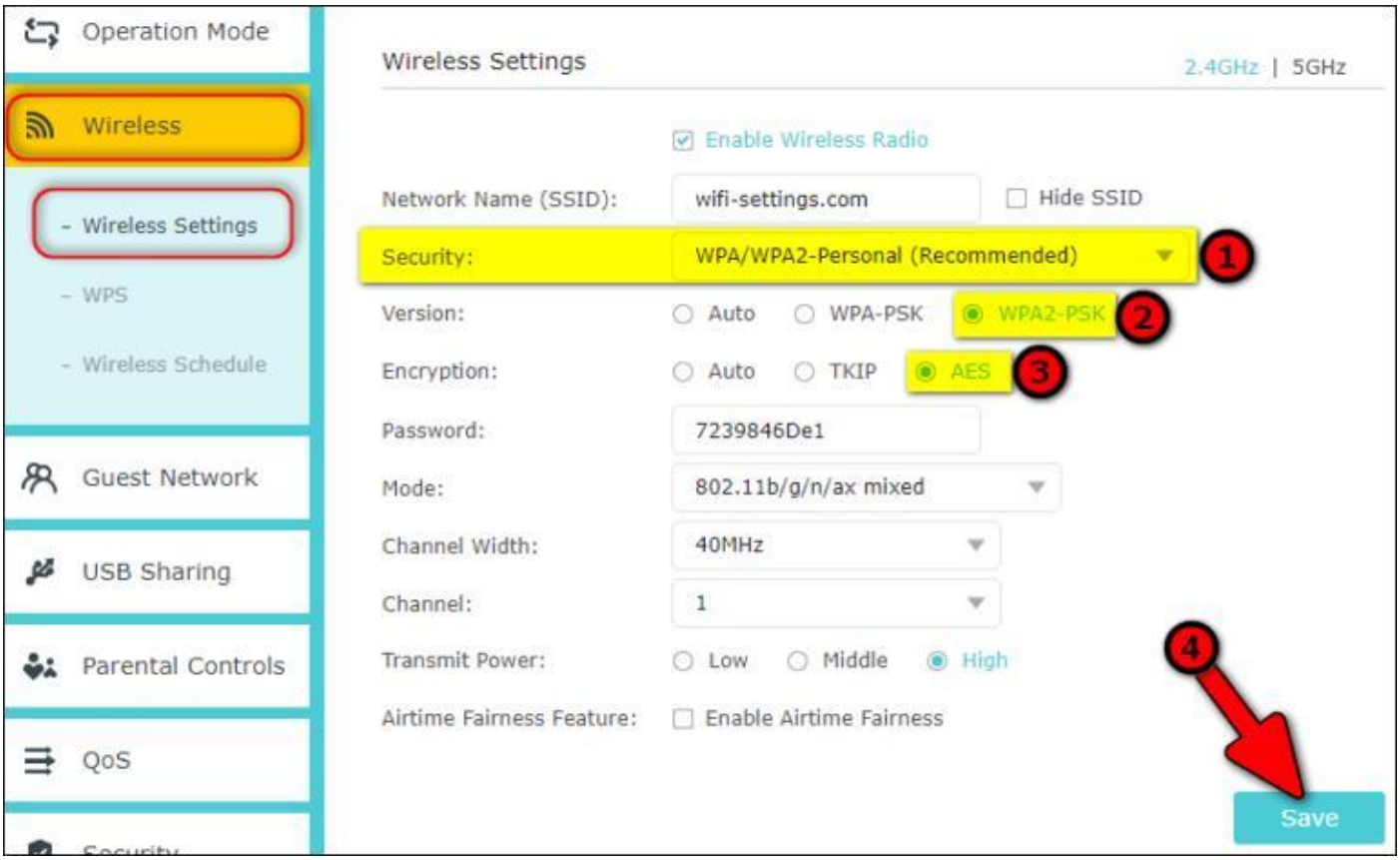
Step 2: Set a Unique SSID



1. **Open Wireless Settings or Wi-Fi Settings.**
2. **Find the SSID / Network Name field.**
3. **Enter a unique name, for example SOHO_Office_WiFi.**
4. **Click Save or Apply.**

SSID means the name displayed when devices search for Wi-Fi networks.

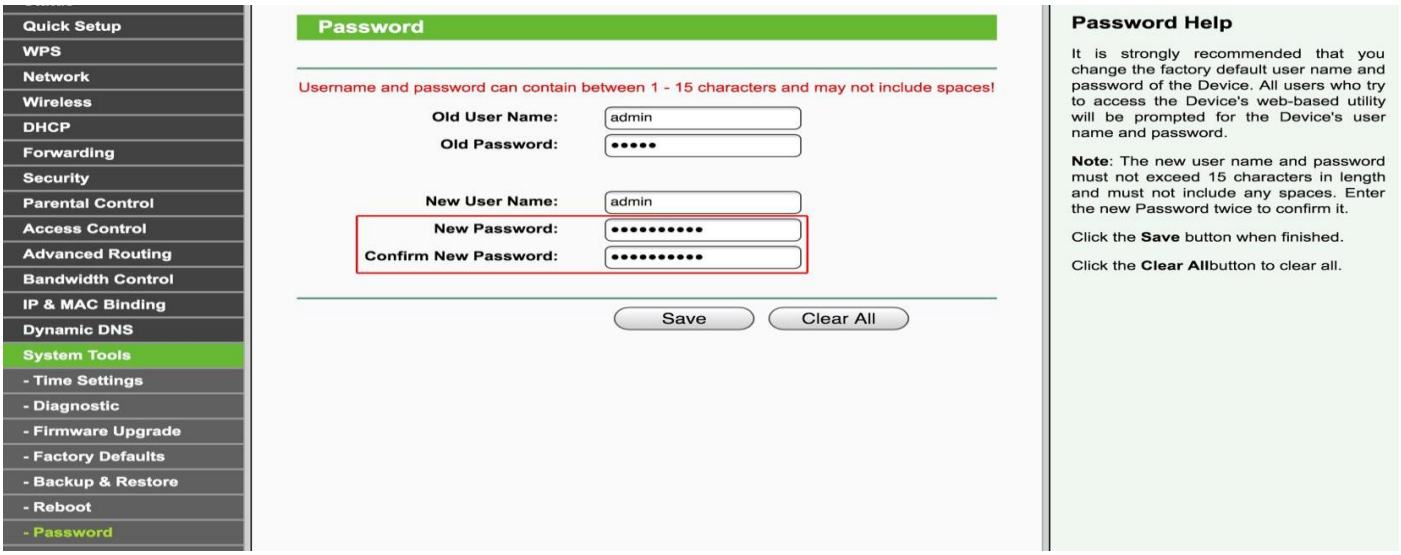
Step 3: Enable WPA2 Security



1. Open Wireless Security settings.
2. Select WPA2-Personal (WPA2-PSK).
3. Choose AES encryption, if available.
4. Enter a strong Wi-Fi password, for example, use a unique password with uppercase and lowercase letters, numbers, and symbols.
5. Click Save or Apply.

WPA2 protects your wireless network from unauthorized access. Use WPA3 if all your devices support it and it is available.

Step 4: Change the Default Admin Password



1. Open Administration, System Tools, or Management.
2. Select Change Password.
3. Enter the current admin password.
4. Set a new, strong administrator password.
5. Confirm the new password and click Save.

Important: The router admin password and Wi-Fi password should be different. Store both securely.

Step 5: Test the Wi-Fi Connection

1. Connect your laptop or phone to SOHO_Office_WiFi.
2. Enter the new Wi-Fi password.
3. Check that the device connects successfully.
4. Open a website to test internet connectivity.
5. Log in to the router using the new admin password to verify it works.

Result: The SOHO Wi-Fi router is configured with a unique SSID, WPA2 security, and a new administrator password.

4. Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

SOHO Network – IP Address Configuration

Network: 192.168.1.0/24

- Printer and Smart TV: Static IP address manually assign in this device.
- Laptops: DHCP through automatically IP address assign.

Device	MAC Address (Example)	IP Address	Type
Wi-Fi Router	00:1A:2B:3C:4D:01	192.168.1.1	Static
Printer	00:1A:2B:3C:4D:02	192.168.1.10	Static
Smart TV	00:1A:2B:3C:4D:03	192.168.1.11	Static
Laptop 1	00:1A:2B:3C:4D:04	192.168.1.100	DHCP
Laptop 2	00:1A:2B:3C:4D:05	192.168.1.101	DHCP
Laptop 3	00:1A:2B:3C:4D:06	192.168.1.102	DHCP

Note: This MAC addresses and IP addresses create an exmple. DHCP ip range 192.168.1.100 between 192.168.1.200 assign a device. then Static IP addresses range out the dhcp range to assing static ip.

5. Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

No.	Possible Cause	Diagnosis and Solution
1	Incorrect Wi-Fi Password	Check the password and reconnect the laptop to the correct Wi-Fi network.
2	Incorrect IP Address	Run `ipconfig` in Command Prompt. Set IP configuration to DHCP and renew the IP using `ipconfig /renew`.
3	Weak Wi-Fi Signal	Move the laptop closer to the router and reconnect to Wi-Fi.
4	Router or Internet Problem	Restart the router and check whether other devices can access the internet.
5	DNS Problem	Run `ipconfig /flushdns` and check whether websites open correctly.

Conclusion: By checking the Wi-Fi password, IP address, signal strength, router, and DNS settings, I can identify and fix most common internet connectivity problems in a SOHO network.